

* Narrow AI - focuses on a single task and is restricted from moving beyond that task to solve unknown problems. General AI can solve many problems using human-like intelligence.

12. Machine learning is a way to train software, called a model, to make predictions or generate content using data.

* Traditional programming is more fixed approach where the programmer designs the solution explicitly, while ML is a more flexible and adaptive approach where ML ^{models} ~~is a more~~ learns from data to generate a solution.

→ Supervised learning - Trains a machine learning algorithm on labelled data sets, effectively giving it a key to understand how to interpret the information it's being fed.

→ Unsupervised learning - Trains the algorithm on unlabeled data, or data without tags.

→ Reinforcement learning - Is the closest type of machine learning to how humans actually learn.

13. Deep learning is a subset of machine learning driven by multilayered neural networks whose design is inspired by the structure of the human brain.

* Deep learning is a subfield of machine learning, machine learning is a subset of AI and AI is the overarching system.

* Neural networks are machine learning algorithms inspired by the human brain.

* Occurs by creating connections and adjusting the weights of connections between neurons through a process called training.

14. Natural Language processing is technology that allows computers to Interpret, manipulate, and Comprehend human language.

- * Auto Correct and Spell Check.

- * Question Answering

- * Machine Translation.

15. Large language model is a type of artificial Intelligence (AI) program that can recognise and generate text among other tasks.

- * Customer Service - large language models provide quick and precise answers to queries from users.

- * Content Creation - They facilitate writing, Summary of text and automatic translation.

16. Generative AI Refers to the use of AI to create new content like text, Images, audio and videos.

→ It is powered by foundation models (large AI models) that can multi-task and perform out of the box tasks.

- * Chat GPT - for versatility

- * Microsoft power apps - for in-depth app development.

- * Adobe photoshop - for AI-generated image enhancement.

17. Training Dataset Refers to the database used to create a representative model by Inferring from the data.

- * High quality data is essential because AI models learn from examples. therefore accurate, diverse and clean data directly determines the models reliability, accuracy and fairness. poor quality data leads to "garbage in, garbage out" Scenarios, resulting in biased, inaccurate and Unreliable predictions that can fail in real-world.

* If it is biased or incorrect, AI models can learn, amplify and solidify those inaccuracies, resulting in discriminating outcomes, unreliable predictions and severe sometimes illegal, real-world consequences.

18 Computer Vision is a field of artificial intelligence (AI) that enables computers and systems to interpret, analyze and derive meaningful information from digital images, videos and other visual inputs.

* By ~~creating~~ converting them into numerical data (pixels) and interpreting patterns using algorithms like Convolutional Neural Networks (CNNs)

→ Retail Shelf monitoring

→ Medical usage Diagnostics.

